22MDC56 – MACHINE LEARNING LABORATORY

PROJECT WORK – PHASE I

Problem Identification and Literature Survey

Bank IQ: A Machine Learning Intelligence Layer for Bank-Statement Analytics — Transaction Classification, Customer Segmentation and Spend Forecasting on a Regulatory Digital Twin

Domain: Machine Learning – Banking / Financial Services Analytics

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Abstract

A bank statement is the most complete record most people have of their own money, yet it answers almost nothing by itself: where did the money go, is this charge normal, what kind of customer am I, what will next month cost. Banks answer these questions internally with hand-written rules — regular expressions over the narration text — which break on the long tail of abbreviations, bank-specific codes and OCR noise, and which cannot learn. Bank IQ is a machine-learning intelligence layer built on top of an existing rule-based system, Vasool Raja — a Regulatory Digital Twin that parses a statement (CSV, PDF or passbook photo), rebuilds the account and runs 19 RBI customer-protection rules to find money the bank owes. The rule engine is kept unchanged; the new layer applies the classical machine-learning syllabus to the same transactions.

The layer is trained on a labelled synthetic dataset of about 130,000 Indian bank-statement lines (500 customers × 12 months, 10 transaction categories, five hidden customer archetypes, seasonality and noisy long-tail narrations), because no public labelled statement data exists. It (1) classifies every line into ten categories with TF-IDF word and character n-grams plus numeric features, comparing Decision Tree, Naive Bayes, kNN and Logistic Regression under stratified 5-fold cross-validation against the regex rule baseline; (2) applies bagging, Random Forest, AdaBoost and gradient boosting; (3) predicts next-month spend from a customer-month panel with linear, regularised, polynomial and tree-based regression under GroupKFold-by-customer cross-validation; (4) segments customers with K-Means and Ward hierarchical clustering, validated by silhouette and Adjusted Rand Index; (5) studies PCA and Truncated SVD; (6) implements locally weighted regression from scratch; and (7) deploys everything as FastAPI endpoints and an Intelligence dashboard in the existing Tamil/English web app, where ML categories, segment, forecast, anomaly flags and plain-language business insights appear beside the rule engine's findings.

Keywords: Transaction Classification, TF-IDF, Naive Bayes, Decision Tree, kNN, Logistic Regression, Random Forest, Gradient Boosting, Regression, Cross-Validation, K-Means, Hierarchical Clustering, PCA, Locally Weighted Regression, Anomaly Detection, Bank Statements, FastAPI

1. Introduction

1.1 Background

Every bank account in India produces a stream of transaction lines — UPI payments, card purchases, ATM withdrawals, transfers, EMIs, charges, salary and pension credits, refunds and interest — each described by a short, cryptic narration such as “UPI/DR/616812345/ANNAPOORNA STORES/CNRB/an***@okaxis” or “CHRG MIN BAL NON MAINT MAY26”. Personal-finance apps, bank analytics teams and consumer-protection tools all need the same first step: turn each line into a meaningful category, then aggregate categories into behaviour — monthly spend by type, income, charges, unusual items — and finally into decisions: which customers behave alike, what should next month look like, what deserves attention.

Vasool Raja, a hackathon project built by the author's team, does the consumer-protection part with deterministic rules: it parses a statement, rebuilds the account as a digital twin and evaluates 19 RBI rules, each at its own effective date, to produce findings with evidence chains. Its transaction classifier is a set of hand-written regular expressions. Those regexes are exact on the formats they were written for and silent on everything else — which, on real statements, is a large share of lines. Machine learning is

the natural next step: learn the categories from labelled examples so that noisy, truncated and bank-specific narrations are handled, and then use the categorised history for segmentation, forecasting and anomaly detection.

1.2 Motivation

The motivation is to apply the full classical machine-learning syllabus — regression with cross-validation; Decision Tree, Naive Bayes and kNN classification; K-Means and hierarchical clustering; dimensionality reduction; bagging, boosting and Random Forest; locally weighted regression; an end-to-end application with a front end — to one real product rather than to seven unrelated toy datasets, and to measure each technique against a meaningful baseline: the regex rules that already exist. The result should be something a user can open, upload a statement to, and read insights from, with the rule engine still the only component that files a claim.

2. Problem Identification

2.1 Problem Statement

Given the transaction lines of a bank account (date, narration, debit, credit, balance) as produced by the Vasool Raja parsers, build, compare and deploy classical machine-learning models that (a) classify each line into one of ten transaction categories more accurately than the existing rule-based classifier, especially on narrations the rules do not cover; (b) predict a customer's total spend in the next month from the current month's behaviour; (c) group customers into behavioural segments without using any demographic attribute; (d) flag unusually large debits in an explainable way; and (e) present the outputs as business insights on a dashboard integrated with the existing application, so that the intelligence layer complements, and never replaces, the rule engine.

2.2 Key Challenges

- No public labelled dataset: bank statements are personal data. A realistic, labelled synthetic dataset must be designed so that labels are not derivable from the narration string alone (otherwise the regex baseline would be trivially perfect).
- Noisy short text: narrations are abbreviated (CHRG, WDL, RVSL), truncated, bank-specific and full of reference numbers; word tokens alone are brittle, so character n-grams and reference-number masking are needed.
- Ten imbalanced classes: bank charges and interest credits are rare but the most important for the product; macro-F1 and per-class F1, not accuracy, must drive model choice.
- Leakage in the regression: a customer's months are correlated, so folds must be grouped by customer (GroupKFold) rather than split at random.
- Honest clustering evaluation: segments must be judged by silhouette and by agreement with a hidden customer type that is never used as a feature.
- Integration: the ML layer must plug into an existing FastAPI service with session authentication and a vanilla-JavaScript bilingual UI without modifying the rule engine.

2.3 Objectives

1. Generate and document a labelled dataset of Indian bank-statement lines with ten categories, five customer archetypes, seasonality and long-tail narrations; perform EDA.

2. Build a preprocessing pipeline: narration cleaning, TF-IDF word (1–2) and character (3–5) n-grams, numeric transaction features; a customer-month panel; a customer behavioural profile.
3. Classify transactions with Decision Tree, Complement Naive Bayes, kNN and Logistic Regression under stratified 5-fold cross-validation, against the regex rule baseline; report accuracy, macro-F1, per-class F1 and confusion matrices.
4. Apply Bagging, Random Forest, AdaBoost and gradient boosting and quantify the gain over a single tree; extract feature importances.
5. Predict next-month spend with Linear, Ridge, Lasso, polynomial, Decision Tree, Random Forest and Gradient Boosting regressors under GroupKFold cross-validation; compare with a naive baseline (RMSE, MAE, R^2).
6. Segment customers with K-Means and Ward clustering; choose k by elbow and silhouette; validate with ARI; name and profile the segments.
7. Apply PCA to the customer profile and Truncated SVD to the text features; implement locally weighted regression from scratch for spend versus income.
8. Deploy the intelligence layer as FastAPI endpoints and an Intelligence dashboard (KPIs, category bars, monthly chart, segment, forecast, anomalies, insights, model card) in the existing Bank IQ web app, with tests, notebooks and Docker.

2.4 Scope

In scope: the ten-category transaction taxonomy, customer-level analytics from a single account's history, the classical models listed above, and the dashboard. Out of scope: deep learning and large language models, fraud detection with labelled fraud data, real customer data (the models are validated on the synthetic test split and demonstrated on the app's synthetic sample statements), and any change to the rule engine, which remains the only source of findings and claims.

3. Existing System and Proposed System

3.1 Existing System

Vasool Raja classifies each transaction with regular expressions in vasool/classify.py (kind and channel), then runs the rulebook. The classifier is deterministic and explainable, but:

- It recognises only the narration formats its authors had seen; merchant-only lines, abbreviations and OCR noise fall to UNKNOWN or to the wrong class. Mapped to the ten ML categories, it reaches about 72% accuracy on the labelled test set.
- It cannot improve from examples; every new bank format needs a new regex.
- It stops at the line: there is no notion of a customer's behaviour, segment, trend or expected spend, and no anomaly detection beyond the rulebook.
- Its outputs are not evaluated: no held-out accuracy, no per-class view of what it misses.

3.2 Proposed System

Bank IQ keeps everything in Vasool Raja and adds a package bankiq/ with a labelled dataset, features, experiments, a service and an API router, plus an Intelligence screen in the web app:

1. Data module: deterministic generator (500 customers × 12 months; ten categories; archetypes salaried / pensioner / student / business / low-activity; seasonality; long-tail templates; noise) → data/ml/transactions.csv; EDA figures.
2. Feature module: narration cleaning with reference-number masking; TF-IDF word + char n-grams; nine numeric features; customer-month panel (34 features incl. per-category amounts/counts and calendar month); 14-feature customer profile.
3. Classification module: regex baseline (rulemap.py) vs Logistic Regression, Complement NB, Decision Tree, kNN; stratified 5-fold CV; sweeps of k and tree depth.
4. Ensemble module: Bagging, Random Forest, AdaBoost, HistGradientBoosting vs single tree; n-gram importances.
5. Regression module: next-month spend; naive baseline; seven regressors; GroupKFold by customer; feature importance.
6. Clustering module: K-Means vs Ward on standardised profiles; elbow, silhouette, dendrogram, ARI vs hidden archetype; segment naming.
7. Dimensionality-reduction and LWR module: PCA on profiles (variance, loadings, segment recovery), Truncated SVD on text, Gaussian-kernel LWR from scratch with a bandwidth study.
8. Service and API: service.analyze(rows) → ML category (+ rule category side by side), category mix, monthly panel, segment, forecast, anomalies, insights; endpoints /api/ml/status, /results, /figures, /analyze, /analyze-json and /api/accounts/{id}/intelligence.
9. Application: Intelligence tab in the existing Tamil/English app; model card with every experiment figure; pytest suite; seven notebooks; Dockerfile and Makefile.

Advantages

- Every category now has a measured accuracy, and the dashboard shows exactly where the learned model and the hand-written rules disagree.
- Customer-level outputs — segment, trend, forecast, anomalies — that the rule engine never had.
- Separation of duties: models describe and predict; rules decide and file claims.
- Reproducible in one command and testable; the whole stack runs on a laptop.

4. Literature Survey

Sixteen references were reviewed in the order the techniques appear in the pipeline: text representation, single-model learners, ensembles, non-parametric regression, clustering and its validation, dimensionality reduction, evaluation protocol, and anomaly detection in finance.

4.1 Induction of Decision Trees

J. R. Quinlan, *Machine Learning*, *vol. 1*, *no. 1*, *pp. 81–106*, *1986.*

Summary: Introduces ID3, which grows a decision tree top-down by choosing at each node the attribute with the highest information gain, and discusses noise, missing values and the over-fitting of fully grown trees.

Relevance to this project: The Decision Tree classifier for transaction categories and the regression tree for next-month spend follow this greedy impurity-driven construction (CART in scikit-learn); the depth sweep in the results reproduces the over-fitting the paper describes.

Limitation: A single tree is unstable — small changes in the data change the tree — which motivates the ensemble methods.

4.2 Nearest Neighbor Pattern Classification

T. Cover and P. Hart, *IEEE Transactions on Information Theory*, *vol. 13*, *no. 1*, *pp. 21–27*, *1967*.

Summary: Proves that the nearest-neighbour rule's error is bounded by twice the Bayes error as the sample grows, establishing kNN as a non-parametric classifier with a strong theoretical guarantee.

Relevance to this project: Justifies kNN as a baseline for narration classification; because it is distance based it runs on the dense SVD-200 projection of the TF-IDF matrix with cosine distance, and the k sweep shows the bias–variance effect of the neighbourhood size.

Limitation: No model is learned, so prediction is O(n) and the method is sensitive to feature scaling, sparsity and irrelevant features.

4.3 Bagging Predictors

L. Breiman, *Machine Learning*, *vol. 24*, *no. 2*, *pp. 123–140*, *1996*.

Summary: Shows that training many versions of an unstable learner on bootstrap resamples and aggregating them (voting or averaging) substantially reduces variance and improves accuracy, especially for decision trees.

Relevance to this project: The Bagging (50 trees) classifier is exactly this recipe and is the first ensemble step above the single tree in the ensemble study.

Limitation: Gains are small for stable learners, and bagged trees remain correlated when a few strong features dominate.

4.4 Random Forests

L. Breiman, *Machine Learning*, *vol. 45*, *no. 1*, *pp. 5–32*, *2001*.

Summary: Extends bagging by also randomising the features considered at each split, which de-correlates the trees; introduces out-of-bag error estimates and variable-importance measures and shows strong results across many datasets.

Relevance to this project: Random Forest is used for transaction classification (on the sparse TF-IDF matrix) and for spend regression; its n-gram importances explain which words the model relies on.

Limitation: Less interpretable than one tree; importance measures are biased toward high-cardinality features.

4.5 A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting

Y. Freund and R. E. Schapire, *Journal of Computer and System Sciences*, *vol. 55*, *no. 1*, *pp. 119–139*, *1997*.

Summary: Introduces AdaBoost, which trains weak learners sequentially, re-weighting misclassified examples so that later learners focus on the hard cases, and proves bounds on training and generalisation error.

Relevance to this project: AdaBoost with depth-2 trees is one of the four ensembles compared; its weaker result on the 10-class narration problem illustrates its sensitivity to noisy, multi-class data.

Limitation: Sensitive to label noise and outliers because misclassified points receive ever-larger weights; slow with many classes.

4.6 Greedy Function Approximation: A Gradient Boosting Machine

J. H. Friedman, *The Annals of Statistics*, *vol. 29*, *no. 5*, *pp. 1189–1232*, *2001*.

Summary: Frames boosting as gradient descent in function space: each new tree fits the negative gradient of the loss, with shrinkage and subsampling for regularisation, giving a general method for regression and classification.

Relevance to this project: Gradient Boosting is the best next-month spend regressor and is the model deployed for the forecast; HistGradientBoosting is one of the classification ensembles.

Limitation: Sequential training is slower than bagging, and depth, learning rate and number of trees interact.

4.7 Robust Locally Weighted Regression and Smoothing Scatterplots

W. S. Cleveland, *Journal of the American Statistical Association*, *vol. 74*, *no. 368*, *pp. 829–836*, *1979*.

Summary: Proposes fitting a low-degree polynomial at each query point by weighted least squares, with weights that decay with distance, and a robustness iteration that down-weights outliers; the bandwidth controls smoothness.

Relevance to this project: The from-scratch Locally Weighted Regression of monthly spend on monthly income implements this idea with a Gaussian kernel and studies the bandwidth τ.

Limitation: Every prediction re-fits a local model (O(n) per query) and the method does not scale to many input dimensions.

4.8 Some Methods for Classification and Analysis of Multivariate Observations

J. MacQueen, *Proc. 5th Berkeley Symposium on Mathematical Statistics and Probability*, *vol. 1*, *pp. 281–297*, *1967*.

Summary: Introduces the k-means procedure that assigns points to the nearest of k centroids and recomputes the centroids until convergence, minimising within-cluster variance.

Relevance to this project: K-Means is the primary customer-segmentation method on the 14-feature behavioural profile; the elbow and silhouette decide k and the clusters are named as business segments.

Limitation: Requires k in advance, assumes roughly spherical clusters and is sensitive to initialisation and scaling.

4.9 Hierarchical Grouping to Optimize an Objective Function

J. H. Ward Jr., *Journal of the American Statistical Association*, *vol. 58*, *no. 301*, *pp. 236–244*, *1963*.

Summary: Describes agglomerative clustering in which the two groups whose merger least increases the total within-group sum of squares are joined at each step, producing a full hierarchy (dendrogram).

Relevance to this project: Ward clustering is compared with K-Means on the same customer profiles; the dendrogram gives a second view of the number of segments and the Adjusted Rand Index measures agreement.

Limitation: $O(n^2)$ memory; merges cannot be undone.

4.10 Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis

P. J. Rousseeuw, *Journal of Computational and Applied Mathematics*, *vol. 20*, *pp. 53–65*, *1987*.

Summary: Defines the silhouette of a point as (b − a)/max(a, b), where a is its mean distance to its own cluster and b to the nearest other cluster, giving a scale-free score in [−1, 1] for comparing clusterings.

Relevance to this project: The silhouette score is the criterion used to choose k for both K-Means and Ward clustering and to compare the two algorithms.

Limitation: O(n^2) pairwise distances; favours compact, well-separated clusters.

4.11 Principal Component Analysis: A Review and Recent Developments

I. T. Jolliffe and J. Cadima, *Philosophical Transactions of the Royal Society A*, *vol. 374*, *no. 2065*, *20150202*, *2016*.

Summary: Reviews PCA as the orthogonal projection maximising retained variance, its computation by eigen-decomposition or SVD, the choice of the number of components, and extensions such as sparse and robust PCA.

Relevance to this project: PCA is applied to the standardised customer profile (variance, loadings, segment recovery vs components) and Truncated SVD plays the same role for the sparse TF-IDF matrix of narrations.

Limitation: Components are linear combinations of all features and can be hard to interpret; variance retained does not guarantee predictive information retained.

4.12 A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection

R. Kohavi, *Proc. 14th International Joint Conference on Artificial Intelligence (IJCAI)*, *pp. 1137–1143*, *1995*.

Summary: Compares hold-out, k-fold, leave-one-out and bootstrap estimators of accuracy on real datasets and recommends stratified k-fold cross-validation as the best trade-off between bias and variance for model selection.

Relevance to this project: All classifier comparisons use stratified 5-fold cross-validation on the training split with a separate held-out test set; the regression uses GroupKFold by customer so that no customer appears in both train and validation folds.

Limitation: Cross-validation estimates have variance themselves and k-fold with small k can be pessimistic.

4.13 Term-Weighting Approaches in Automatic Text Retrieval

G. Salton and C. Buckley, *Information Processing & Management*, *vol. 24*, *no. 5*, *pp. 513–523*, *1988*.

Summary: Compares term-frequency, inverse-document-frequency and length-normalisation schemes for representing text as weighted term vectors and shows that tf·idf with cosine normalisation gives the best retrieval performance.

Relevance to this project: Every bank-statement narration is turned into a TF-IDF vector (word 1–2-grams and character 3–5-grams) before classification; the weighting lets rare, informative tokens (SALARY, NACH, REV) outweigh frequent reference tokens.

Limitation: Bag-of-words ignores word order and semantics; vocabulary grows with the corpus.

4.14 Tackling the Poor Assumptions of Naive Bayes Text Classifiers

J. D. M. Rennie, L. Shih, J. Teevan and D. R. Karger, *Proc. 20th International Conference on Machine Learning (ICML), pp. 616–623, 2003.*

Summary: Identifies why multinomial Naive Bayes under-performs on text (skewed class sizes, weight magnitude errors) and proposes Complement Naive Bayes with TF-IDF transforms and weight normalisation, which closes most of the gap to SVMs.

Relevance to this project: Complement Naive Bayes is the Naive Bayes variant used for the 10-class narration problem, because class sizes are imbalanced and the features are TF-IDF counts rather than Gaussian measurements.

Limitation: Still assumes conditional independence of terms; calibration of probabilities is poor.

4.15 N-Gram-Based Text Categorization

W. B. Cavnar and J. M. Trenkle, *Proc. 3rd Annual Symposium on Document Analysis and Information Retrieval (SDAIR), pp. 161–175, 1994.*

Summary: Represents documents by their character n-gram frequency profiles and shows that such profiles are robust to spelling errors, truncation and OCR noise, giving reliable language and topic categorisation.

Relevance to this project: Bank narrations are abbreviated, truncated and noisy (CHRG, WDL, RVSL, OCR from passbooks); character 3–5-grams are what let the classifier recover meaning that word tokens and hand-written regexes miss.

Limitation: Large, redundant feature spaces; n-grams carry no semantics beyond surface form.

4.16 A Survey of Anomaly Detection Techniques in Financial Domain

M. Ahmed, A. N. Mahmood and M. R. Islam, *Future Generation Computer Systems, vol. 55, pp. 278–288, 2016.*

Summary: Surveys statistical, clustering-based and classification-based anomaly-detection methods applied to financial data, with the taxonomy of point, contextual and collective anomalies and a discussion of the scarcity of labelled data.

Relevance to this project: Bank IQ's unusual-debit flag is a contextual point anomaly: a robust z-score of the amount within its ML category. The survey justifies a transparent statistical detector where labelled fraud data is unavailable.

Limitation: Survey only; no benchmark on bank-statement data; most methods need tuning per dataset.

5. Comparative Analysis of Surveyed Literature

No.	Paper (Year)	Technique	Strength	Limitation
1	Induction of Decision Trees (1986)	Decision tree induction (ID3)	Interpretable rules; handles mixed features	Unstable; over-fits without pruning
2	Nearest Neighbor Pattern Classification (1967)	k-nearest neighbours	Simple, non-parametric, strong asymptotic bound	Slow at prediction; weak on sparse text
3	Bagging Predictors (1996)	Bootstrap aggregating	Variance reduction for unstable learners	Trees stay correlated
4	Random Forests (2001)	Random forest	Robust, de-correlated trees, built-in importance	Black-box; importance bias
5	A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting (1997)	AdaBoost	Sequential focus on hard examples	Sensitive to noise; slow multi-class
6	Greedy Function Approximation (2001)	Gradient boosting	State-of-the-art on tabular data	Slower; several interacting hyperparameters
7	Robust Locally Weighted Regression and Smoothing Scatterplots (1979)	Locally weighted regression (LOWESS)	Flexible non-parametric fit	O(n) per prediction; low-dimensional
8	Some Methods for Classification and Analysis of Multivariate Observations (1967)	k-means clustering	Fast, scalable segmentation	k chosen by hand; spherical clusters
9	Hierarchical Grouping to Optimize an Objective Function (1963)	Agglomerative (Ward) clustering	No k needed up front; full hierarchy	$O(n^2)$ memory; greedy merges
10	Silhouettes (1987)	Silhouette score	Scale-free cluster quality score	Favours compact clusters
11	Principal Component Analysis (2016)	PCA / SVD	Compact, decorrelated representation	Harder to interpret
12	A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection (1995)	Cross-validation	Reliable model selection protocol	Estimates still have variance
13	Term-Weighting Approaches in Automatic Text Retrieval (1988)	TF-IDF term weighting	Simple, strong text representation	No word order or semantics
14	Tackling the Poor Assumptions of Naive Bayes Text Classifiers (2003)	Complement Naive Bayes	Fast NB that handles class imbalance	Independence assumption; poor calibration
15	N-Gram-Based Text Categorization (1994)	Character n-gram features	Robust to noise, truncation, OCR	Large feature space
16	A Survey of Anomaly Detection Techniques in Financial Domain (2016)	Anomaly detection in finance	Taxonomy and methods for financial anomalies	No ready benchmark; per-dataset tuning

Table 5.1: Comparison of surveyed literature

6. Research Gap and Contribution

The surveyed methods are mature individually: text can be represented with TF-IDF and character n-grams [13], [15] and classified with Naive Bayes [14], trees [1], neighbours [2] or ensembles [3]–[6]; customers can be clustered [8]–[10] and compressed [11]; forecasts can be fitted globally or locally [7]; evaluation has a standard protocol [12]; and financial anomaly detection has a taxonomy [16]. What is missing is their combination on bank-statement data with an honest baseline:

1. Published transaction-categorisation work uses proprietary bank data and is rarely compared with the rule systems banks actually run; here the regex classifier is the measured baseline and the per-class gain is reported.
2. Classification, segmentation, forecasting and anomaly detection are treated as separate problems; here they are built on one feature pipeline from one account history, so the segment, the forecast and the flags are consistent with the categories.
3. Nothing in the literature is integrated into a consumer-protection tool; here the ML layer sits beside a deterministic rule engine, with a clear rule that models inform and rules decide.
4. No public labelled statement dataset exists; the generator in this project is documented and reproducible, with labels independent of the narration string, so the evaluation is meaningful.

Bank IQ contributes (i) a reproducible labelled dataset and feature pipeline for Indian bank narrations, (ii) a side-by-side evaluation of the syllabus techniques against a real rule baseline, (iii) customer-level analytics with cross-validation that respects customer grouping, and (iv) a deployed intelligence dashboard inside an existing bilingual application.

7. Tools and Technologies

Layer	Technology	Purpose
Language	Python 3.11	All code
Data	pandas, NumPy	Generator, panel, profile
Text features	scikit-learn TfidfVectorizer (word + char_wb), TruncatedSVD	Narration representation
Models	scikit-learn: LogisticRegression, ComplementNB, DecisionTree, kNN, Bagging, RandomForest, AdaBoost, HistGradientBoosting, Linear/Ridge/Lasso, GradientBoostingRegressor, KMeans, AgglomerativeClustering, PCA	All syllabus models
From scratch	NumPy	Locally weighted regression
Visualisation	matplotlib, SciPy dendrogram, inline SVG in the UI	15 result figures + dashboard charts
Application	FastAPI, vanilla JavaScript (existing app), joblib	Endpoints, Intelligence screen, saved models
Engineering	pytest, Docker, Makefile, Jupyter	Tests, deployment, reproducibility, notebooks

Table 7.1: Tools and technologies

8. Work Plan

Phase	Component	Deliverables	Schedule
I	Problem identification and literature survey	This document; summary of research articles	16 Sep 2026
II	User interface and backend connectivity	Intelligence screen ↔ /api/ml ↔ saved models	30 Sep 2026
II	Preprocessing and EDA	Generator, cleaning, features, panel, profile; EDA notebook and figures	30 Sep 2026
II	Model building	Classification, ensembles, regression, clustering, PCA/SVD, LWR	30 Sep 2026
II	Result analysis and business insights	Cross-validated tables, figures, insights engine	30 Sep 2026
III	Deployment and demo	Docker, tests, demo script	CAT II
III	Presentation and viva	Slides, viva sheet	CAT II
III	Report and GitHub	Project report, repository	CAT II

Table 8.1: Phase-wise work plan

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